

[134]

SARDAR PATEL UNIVERSITY

US04CBCA52 – Data Structure - II

SYBCA (Sem-4)

Date: 03.04.2025

Time: 02.00 PM - 05.00 PM

Marks: 70



Q-1 MCQ

[10]

1. The size of one dimensional array can be calculated by using ____ formula.
A. $2^{d+1} - 1$
B. $2^{d+1} + 1$
C. $2^{d-1} - 1$
D. $2^{d-1} + 1$
2. A binary tree has at most _____ child.
A. One
B. Two
C. Three
D. None of the above
3. Each node in singly linked list has _____ fields.
A. 1
B. 2
C. 3
D. 4
4. A data structure that contains not only a data field but also contains pointer field is known as ____
A. Stack
B. Queue
C. Tree
D. Linked List
5. ____ is the process to arrange the data in ascending or descending order.
A. Sorting
B. Inserting
C. Searching
D. Updating
6. Quick sort is also known as...
A. merge sort
B. heap sort
C. bubble sort
D. None of these
7. The lowest level of index is _
A. Track index
B. Index area
C. Prime index
D. Master index
8. _____ file is also known as Random file.
A. Serial
B. Sequential
C. Direct
D. None of the above
9. Leaf node is known as _____ node.
A. Terminal
B. Parent
C. Root
D. None of the above
10. Merge sort uses...
A. Divide and Conqueror
B. Dynamic Programming
C. Greedy
D. Backtracking

C.P. T.O.)

Q-2 Give answer in short. (ANY TEN)

[20]

1. Define with an example: Root node and Level of a tree.
2. List out Applications of Tree.
3. Define Traversals. List the types of binary tree traversal
4. List insert operations that can be performed on singly linked list.
5. What is a linked list? List out their types.
6. Explain singly linked list in brief.
7. Define sorting. List out the names of sorting techniques.
8. List out application of sorting.
9. Define searching. List out the names of searching techniques.
10. List popular hashing functions.
11. Define: Hashing function, Collision resolution technique
12. List types of Chaining.

Q-3 A. Write a note on the memory representation of a binary tree using sequential method. **[5]**

B. Write down an algorithm for inorder traversal of a binary tree. **[5]**

OR

Q-3 A. Write a note on the memory representation of a binary tree using Linked method. **[5]**

B. Explain the insertion of a node in a binary with an example. **[5]**

Q-4 A. Write an algorithm to insert node at front position in singly linked list. **[5]**

B. Write an algorithm to delete node from end position in singly linked list. **[5]**

OR

Q-4 A. Write an algorithm to insert node at end position in singly linked list. **[5]**

B. Write an algorithm to delete node at beginning in singly linked list. **[5]**

Q-5 A. Write down the algorithm of Bubble sort. **[5]**

B. Write down the algorithm of Linear Search. **[5]**

OR

Q-5 A. Write down the algorithm of Quick sort. **[5]**

B. Differentiate between Sorting and Searching. **[5]**

Q-6 A. Explain ANY FOUR hashing functions with example. **[10]**

OR

Q-6 A. Write a detail note on index sequential file supported by IBM **[10]**

-----X-----